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Education

Netaji Subhash University of Technology

Electrical Engineering (B.Tech)

August 2021 – May 2025

Delhi, India

Projects

CIBMTR Survival Prediction — Machine learning task | [Link](#)

- Achieved **global rank of 97th** out of **3,325 teams** and won a **silver medal** in the CIBMTR Kaggle competition
- The competition aimed to predict survival rates post-Hematopoietic Cell Transplant (HCT) using C-index metric
- Built **multiple-seed GBDT models** (CatBoost, XGBoost, LightGBM) for Kaplan-Meier, Cox, and Nelson-Aalen estimators with a CV score of **0.683**
- Leveraged a public **neural network** with log-time transformation, achieving a CV score of **0.6825**, and combined it with GBDT models to achieve a final CV score of **0.685**

Meesho Fashion Attribute Classification — Computer Vision Task | [Link](#)

- Ranked **11th out of 196** teams in the Meesho ML Challenge by developing a model for feature prediction from product images
- Addressed missing values in the Sarees category using **EasyOCR** and product ID grouping, improving the score
- Generated 5 distinct augmented image embeddings from **CLIP and ViT models**, and utilized them with an **LGBM classification model** for each product category and attribute
- Employed feature removal, early stopping, and **5-stratified k-fold cross-validation** to optimize the final score

Crab Age Prediction — Machine learning task | [Link](#)

- Applied 3 models (**XGBoost, CatBoost, LightGBM**) and ensembled them to achieve a private dataset score of **1.33589, ranking 6th out of 1431 teams**
- Improved MAE by **20-25%** through outlier removal and data cleaning of over **40,000 data points**, leading to enhanced model stability and the most significant performance boost
- Integrated additional data sources, enhancing training size and improving prediction accuracy, which contributed to **10-15%** better generalization
- Reduced overfitting by **20%** through **advanced ensembling techniques**, combining predictions from multiple models for the final submission

Student Essay Score Prediction — Natural Language Processing task | [Link](#)

- Created a machine learning model for the Learning Agency Lab - Automated Essay 2.0 Kaggle competition
- Fine-tuned **DeBERTa** (Decoding-enhanced BERT with Disentangled Attention) and generated out-of-fold predictions through **stratified K-fold cross-validation**, achieving a **0.805 QWK** score
- Combined essay-specific features with out-of-fold (OOF) predictions, integrating both into **Gradient Boosting Decision Tree (GBDT) models** to achieve a score of **0.816**, and further ensembled predictions to derive final essay scores
- Attained a best private score of **0.835** using the competition's quadratic weighted Kappa metric

Technical Skills

Languages: Python, C/C++, SQL

Frameworks: PyTorch, TensorFlow, Scikit-Learn, Node.js, React, Express, MongoDB

Development: Machine learning, Software Development, Full Stack development

Version Control: Git, Github

CS Fundamentals: Operating System, DataBase Management System, Computer Networks, System Design

Achievements

- Secured Knight rank on [LeetCode](#) (Top 2% globally) with a competition **rating exceeding 2100**
- Attained Specialist rank on [CodeForces](#) with a **peak rating of 1561**, ranking in the **Top 2000 in India** with strong competitive programming skills
- Placed **3rd** in IIT BHU Vista Hackathon on [Kaggle](#) for AI generated image detection